Multiplicative GARCH-X with VIX: A Parsimonious Alternative to GARCH-MIDAS for Volatility Forecasting and Risk Management*

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Abstract

We propose a daily multiplicative GARCH-X model, $\sigma_t^2 = \tau_t \times g_t$, where τ_t is a simple function of lagged VIX and g_t follows a GJR-GARCH process on standardized returns. In a systematic horse race among 17 specifications—including ten daily GARCH-X variants, six GARCH-MIDAS alternatives with Beta-polynomial weighting, and a GJR-GARCH benchmark—the simplest daily model with $\tau_t = \theta_0 + \theta_1 \text{VIX}_{t-1}^2$ and freely estimated intercept (A4f) achieves the lowest QLIKE loss function, significantly outperforming GJR-GARCH (Diebold-Mariano $t = 4.03$, surpassing the Harvey et al. (2016) threshold of $|t| > 3.0$) and outperforming all GARCH-MIDAS specifications in point estimates. The result extends to QQQ ($t = 3.71$), European equities (EURO STOXX 50: $t = 3.64$; FEZ: $t = 3.45$), and, when asset-specific fear indices are used, to GLD with GVZ ($t = 3.17$). We show that the g_t component *contemporaneously* tracks the variance risk premium (Spearman $\rho \approx 0.80$), providing a structural interpretation: τ_t reflects the option-market fear level while g_t captures deviations of realized variance from implied variance.

*All errors are my own. Data and replication code are available upon request.

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Importantly, g_t has no out-of-sample *predictive* power for future VRP—it is a decomposition tool, not a forecasting signal. Residual diagnostics reveal that VIX absorption reduces excess kurtosis by 60% and shifts optimal Student-t degrees of freedom from 5.3 to 8.0, directly explaining the improved tail risk calibration. For risk management, A4f with Student-t innovations passes VaR backtesting at two of three confidence levels (scorecard 3/4), whereas GJR-GARCH passes none (scorecard 1/4). These findings demonstrate that a parsimonious multiplicative GARCH-X specification with daily VIX is statistically indistinguishable from the best GARCH-MIDAS alternative (B1, $K = 22$; DM $t = -0.90$) while significantly dominating longer-lag MIDAS specifications ($|t| > 3.0$), delivering both superior point forecasts and improved tail risk measurement.

Keywords: Volatility forecasting, GARCH-X, GARCH-MIDAS, VIX, variance risk premium, source decomposition, Value-at-Risk, Expected Shortfall

JEL Classification: C22, C53, G12, G17

Contents

1 Introduction

Accurately forecasting equity return volatility is central to risk management, derivative pricing, and portfolio construction. A large literature documents that the CBOE Volatility Index (VIX), derived from S&P 500 index option prices, is a powerful—though upward-biased—predictor of future realized volatility (Christensen and Prabhala, 1998; Jiang and Tian, 2005; Bekaert and Hoerova, 2014). Yet the question of how best to integrate VIX information into parametric volatility models remains open.

One influential approach is the GARCH-MIDAS framework of Engle et al. (2013), which decomposes conditional variance into a slowly moving long-run component τ_t and a mean-reverting short-run component g_t through a multiplicative structure $\sigma_t^2 = \tau_t \times g_t$. The long-run component is modeled as a mixed-data-sampling (MIDAS) filter that aggregates lagged values of a lower-frequency variable—such as realized volatility, macroeconomic indicators, or financial conditions indices—using Beta-polynomial weights. This framework has been extended by Conrad and Loch (2015) to incorporate forward-looking variables and by Conrad and Kleen (2020) to include multiple external regressors.

However, when the external variable of interest is VIX, which updates at daily frequency, the MIDAS mixed-frequency machinery becomes conceptually redundant. VIX is already a daily variable; aggregating it through Beta-polynomial weights over 22, 65, or 125 lags introduces additional parameters without a clear economic rationale. This observation motivates a fundamental question: *can a simple daily multiplicative model with VIX match or exceed the forecasting performance of the more complex GARCH-MIDAS specification?*

We address this question through three contributions.

First, we conduct the most comprehensive specification comparison to date of 17 multiplicative volatility models, encompassing ten daily GARCH-X variants that differ in the functional form of τ_t (log-exponential, VIX-squared, VIX-level), the choice of normalization denominator (τ_t versus τ_{t-1}), and the treatment of the g_t intercept (constrained to enforce $\mathbb{E}[g_t] = 1$ versus freely estimated), alongside six GARCH-MIDAS specifications with rolling-window and fixed-span Beta-polynomial filters at various lag lengths, and a standard GJR-GARCH(1,1) benchmark. Using S&P 500 daily returns from 2005 to 2026 with an out-of-sample (OOS) period spanning 2019–2026 (1,825 observations), we evaluate all models under the proxy-robust QLIKE loss function of Patton (2011) and the stringent Diebold-Mariano testing framework with the Harvey et al. (2016) multiple-testing threshold of $|t| > 3.0$. The simplest daily model—$\tau_t = \theta_0 + \theta_1 \mathrm{VIX}^2_{t-1}$ with freely estimated g_t intercept (denoted A4f)—achieves the lowest QLIKE and a DM test statistic of $t = 4.03$ against GJR-GARCH. While the Model Confidence Set cannot distinguish A4f from other well-specified VIX models (all survive at $\alpha = 0.10$), A4f and a majority of the VIX-augmented specifications clear the $|t| > 3$ screen against plain GJR-GARCH; the weaker variants do not. A4f remains the point-estimate best among functionally equivalent alternatives. Notably, GARCH-MIDAS specifications with longer lag structures ($K = 65$–125) are significantly inferior to A4f ($|t| > 3.0$), while the shortest-lag MIDAS (B1, $K = 22$) is statistically indistinguishable (DM $t = -0.90$)—demonstrating that parsimony, not statistical dominance over all MIDAS variants, characterizes the advantage of daily GARCH-X when VIX updates daily.

Second, we propose a *source decomposition* interpretation that differs from the standard long-run/short-run narrative. In our framework, τ_t captures the *exogenous* volatility level implied by the options market (the "fear level"), while g_t captures *endogenous* GARCH dynamics conditional on this external environment. We demonstrate that g_t tracks the variance risk premium (VRP) with Spearman rank correlation $\rho \approx 0.80$—far exceeding the raw $r_t^2/\mathrm{VIX}^2_{t-1}$ ratio ($\rho = 0.15$)—indicating that the GARCH autoregressive structure substantially enhances VRP measurement. However, we also report the honest null result that g_t possesses no exploitable out-of-sample predictive power for future VRP, consistent with the high noise level and low autocorrelation of daily VRP.

Third, we validate the model's practical value through VaR and Expected Shortfall (ES) backtesting. A4f with Student-t innovations achieves a scorecard of 3/4 (passing VaR at 1% and 5% levels plus ES at 2.5%), while GJR-GARCH achieves only 1/4 regardless of distributional assumption. Cross-asset validation shows that the model generalizes to assets with high VIX correlation (QQQ: DM $t = 3.71$) and, when asset-specific implied volatility indices are employed, to gold (GLD with GVZ: DM $t = 3.17$). For assets with weak VIX linkage—emerging markets (EEM) and Taiwan (0050.TW)—the model improves directionally but does not reach statistical significance, consistent with the interpretation that the multiplicative structure requires a relevant fear index as input.

Our findings contribute to the literature on component volatility models (Engle and Rangel, 2008; Engle et al., 2013; Conrad and Loch, 2015; Conrad and Kleen, 2020), the role of VIX in volatility forecasting (Bekaert and Hoerova, 2014), and the practical evaluation of volatility models for risk management (Patton, 2011; Hansen et al., 2011). The key message is one of parsimony: when the external variable is already daily, GARCH-MIDAS specifications with long lag structures ($K = 65$–125) are significantly dominated by a simple daily multiplicative model ($|t| > 3.0$), while the shortest-lag alternative (B1, $K = 22$) is statistically indistinguishable—parsimony rather than statistical dominance is the practical argument for the daily specification. A simple multiplicative structure with daily VIX delivers both superior forecasting and improved risk management without the added complexity of mixed-frequency aggregation.

The remainder of this paper is organized as follows. Section 2 reviews the related literature. Section 3 presents the model specifications and evaluation framework. Section 4 describes the data. Section 5 reports the empirical results. Section 6 presents robustness analyses. Section 8 concludes.

2 Related Literature

2.1 Component Volatility Models

The idea of decomposing conditional variance into multiple components has a rich history. Engle and Rangel (2008) introduced the Spline-GARCH model, where a slowly moving spline function captures the unconditional variance trend while a unit-GARCH process captures short-run dynamics. Engle et al. (2013) generalized this idea with the GARCH-MIDAS framework, replacing the spline with a MIDAS filter that aggregates lagged values of observable variables—realized volatility, macroeconomic indicators, or financial variables—through Beta-polynomial weights. The appeal of GARCH-MIDAS lies in its ability to incorporate lower-frequency information (e.g., quarterly GDP) into a daily volatility model without frequency mismatch.

Subsequent work has expanded the framework in several directions. Conrad and Loch (2015) incorporated forward-looking variables, showing that consumer confidence and financial conditions indices improve long-horizon volatility forecasts. Conrad and Kleen (2020) demonstrated that combining realized volatility with macroeconomic variables in a two-component MIDAS filter outperforms single-variable specifications. Wang and Ghysels (2015) compared GARCH-MIDAS with alternative component models and found that the choice of explanatory variable matters more than the component structure itself.

However, a limitation of existing GARCH-MIDAS studies is that they focus on mixed-frequency settings where the MIDAS aggregation is economically motivated (e.g., monthly macro data driving daily volatility). When the external variable is itself daily—as is the case with VIX—the longer-lag MIDAS filter adds parameters without measurable forecasting benefit. Our paper fills this gap by showing that a simple daily multiplicative model significantly outperforms longer-lag MIDAS alternatives ($K \geq 65$) while achieving statistically indistinguishable performance relative to the shortest-lag MIDAS (B1, $K = 22$).

2.2 VIX and Volatility Forecasting

The role of VIX in volatility forecasting has been extensively studied. Christensen and Prabhala (1998) showed that implied volatility subsumes the information content of historical volatility. Jiang and Tian (2005) demonstrated that model-free implied volatility is a more efficient forecast than Black-Scholes implied volatility. Bekaert and Hoerova (2014) decomposed VIX^2 into expected variance and a variance risk premium component, showing that both contribute to volatility dynamics.

Several papers have incorporated VIX directly into GARCH models. Han and Kristensen (2014) developed asymptotic theory for GARCH-X models, establishing consistency and asymptotic normality of quasi-maximum likelihood estimators when exogenous regressors are added to the variance equation. Francq and Thieu (2019) extended this framework to allow for possibly nonstationary covariates, which is relevant given VIX's occasional extreme persistence during crisis periods. Their theoretical results justify the inclusion of VIX in GARCH specifications under mild regularity conditions. The general finding in this literature is that simple approaches to incorporating VIX often perform as well as complex ones—a result consistent with our findings.

Our contribution relative to this literature is twofold. First, we use a *multiplicative* rather than additive specification for incorporating VIX, which has the advantage of scaling the entire variance process rather than simply adding a level term. Second, we systematically compare the functional form of the VIX component (τ_t), showing that the VIX-squared specification dominates alternatives.

2.3 Variance Risk Premium

The variance risk premium—defined as the difference between risk-neutral and physical expected variance—has attracted substantial attention since Bollerslev et al. (2009) showed that VRP predicts aggregate stock returns. Carr and Wu (2009) provided a theoretical framework for variance swaps, and Bekaert and Hoerova (2014) decomposed VIX^2 into expected variance and VRP.

Our paper connects to this literature through the g_t component. When $\tau_t \approx \text{VIX}^2_{t-1}$, the ratio $g_t \approx \sigma^2_t/\text{VIX}^2_{t-1}$ approximates the realized-to-implied variance ratio, which is inversely related to VRP. We show that the GARCH autoregressive structure in g_t substantially improves VRP tracking relative to the raw ratio, but that this improvement does not translate into out-of-sample VRP predictability.

3 Methodology

3.1 The Multiplicative GARCH-X Framework

We specify the daily return r_t of an equity index as:

$$r_t = \mu + \sigma_t \varepsilon_t, \qquad \varepsilon_t \sim D(0,1), \tag{1}$$

where μ is the unconditional mean return, σ_t^2 is the conditional variance, and ε_t is an i.i.d. innovation from distribution D with zero mean and unit variance. We set $\mu = 0$ throughout, which is standard for daily data where the mean is negligible relative to the variance.

The conditional variance, modeled within the GARCH framework (Engle, 1982; Bollerslev, 1986), follows a multiplicative decomposition:

$$\sigma_t^2 = \tau_t \times g_t, \tag{2}$$

where $\tau_t > 0$ is the *exogenous scale factor* determined by the options-implied fear level (VIX), and $g_t > 0$ is the *endogenous dynamic factor* that captures GARCH-type persistence conditional on the external environment.

3.1.1 Scale Factor Specifications

We consider five functional forms for τ_t, all based on the lagged VIX to avoid look-ahead bias:

(a) **Log-exponential** (baseline):

$$\tau_t = \exp(\theta_0 + \theta_1 \log \mathrm{VIX}_{t-1}). \tag{3}$$

(b) **VIX-squared** (variance-matching):

$$\tau_t = \max\big(\theta_0 + \theta_1 \mathrm{VIX}_{t-1}^2, \epsilon\big), \tag{4}$$

where $\epsilon = 10^{-16}$ is a positivity floor. Since VIX is quoted in annualized volatility units (%), VIX^2 has variance units, matching the dimension of σ_t^2. We use VIX in its raw form (annualized percentage); the coefficients θ_0 and θ_1 absorb the necessary unit conversions to daily decimal variance.

(c) **VIX-level** (exponential):

$$\tau_t = \exp(\theta_0 + \theta_1 \,\text{VIX}_{t-1}). \tag{5}$$

(d) **Sample-mean normalized**: Replace the intercept constraint with division by the in-sample mean of τ_t, so that $\mathbb{E}[\tau_t] = 1$ by construction:

$$\tau_t^{(n)} = \frac{\tau_t}{\bar{\tau}_{\text{train}}}, \qquad g_t^{(n)} = \bar{\tau}_{\text{train}} \times g_t. \tag{6}$$

3.1.2 Dynamic Factor Specification

The short-run component g_t follows a GJR-GARCH process (Glosten et al., 1993) on standardized returns:

$$g_t = \omega_g + \alpha\, u_{t-1}^2 + \gamma\, u_{t-1}^2 \mathbb{1}_{[u_{t-1}<0]} + \beta\, g_{t-1}, \tag{7}$$

where the standardized return u_{t-1} is defined under one of two normalization conventions:

$$u_{t-1}^{(a)} = \frac{r_{t-1}}{\sqrt{\tau_t}} \qquad \text{(contemporaneous normalization)}, \tag{8}$$

$$u_{t-1}^{(b)} = \frac{r_{t-1}}{\sqrt{\tau_{t-1}}} \qquad \text{(lagged normalization)}. \tag{9}$$

Contemporaneous normalization follows the recursive structure of Engle et al. (2013, Eq. 4), where the current volatility environment determines how past shocks are interpreted.[1] Despite the shared time subscript, there is no simultaneity: τ_t is a deterministic

[1] This timing convention is a design choice rather than an identification necessity. We explicitly compare the Engle-style recursion $u_{t-1} = r_{t-1}/\sqrt{\tau_t}$ against the more conservative DGP-timed alternative $u_{t-1} = r_{t-1}/\sqrt{\tau_{t-1}}$. The main paper adopts the former as its baseline because τ_t is predetermined by VIX_{t-1} and therefore known before r_t is realized. A later source-code audit of downstream robustness

function of VIX_{t-1}, which is observed before r_t. Lagged normalization reflects a conservative timing choice in which the normalization denominator τ_{t-1} is fully predetermined before the shock r_{t-1} is observed, since both τ_t and u_{t-1} use VIX_{t-1} information. Our baseline specifications use contemporaneous normalization; lagged normalization is included for comparison (see Section 3.2).

Two intercept treatments are considered:

(i) **Constrained** ($\mathbb{E}[g_t] = 1$): Setting $\omega_g = 1 - \alpha - \gamma/2 - \beta$ forces the unconditional mean of g_t to unity, so that $\mathbb{E}[\sigma_t^2] = \mathbb{E}[\tau_t]$ and the scale factor τ_t determines the unconditional variance level.

(ii) **Free**: ω_g is estimated without constraint. This allows $\mathbb{E}[g_t] \neq 1$, providing an additional degree of freedom that can improve fit when VIX systematically over- or under-estimates realized variance (i.e., when a non-zero VRP is present).

3.2 Specification Taxonomy

Table 1 presents the full taxonomy of 17 specifications examined in this paper.

scripts found that some follow-up experiments inherited this Engle-style recursion without always stating the convention explicitly; a dedicated refit confirms that the directional conclusions are preserved under the lagged-normalization alternative, although some magnitudes shift. We therefore interpret the τ_t baseline as an implementation convention inherited from Engle et al. (2013), not as the only admissible timing assumption.

Table 1: Taxonomy of 17 multiplicative volatility model specifications

#	Name	τ_t function	u_{t-1} denom.	ω_g	Category
A1	Original	log-exp	τ_t (est) / τ_{t-1} (OOS)	constrained	GARCH-X
A2	consistent-τ_t	log-exp	τ_t	constrained	GARCH-X
A3	consistent-τ_{t-1}	log-exp	τ_{t-1}	constrained	GARCH-X
A4	VIX-squared	VIX^2	τ_t	constrained	GARCH-X
A5	VIX-level	exp(VIX)	τ_t	constrained	GARCH-X
A2f	free-omega	log-exp	τ_t	free	GARCH-X
A4f	VIX^2-free	VIX^2	τ_t	free	GARCH-X
A3f	τ_{t-1}-free	log-exp	τ_{t-1}	free	GARCH-X
A2n	log-exp-norm	log-exp, norm	τ_t	constrained[a]	GARCH-X
A4n	VIX^2-norm	VIX^2, norm	τ_t	constrained[a]	GARCH-X
B1	MIDAS-RW-K22	Beta poly, $K = 22$	τ_t	constrained	MIDAS-RW
B2	MIDAS-RW-K65	Beta poly, $K = 65$	τ_t	constrained	MIDAS-RW
B3	MIDAS-RW ($K = 125$)	Beta poly, $K = 125$	τ_t	constrained	MIDAS-RW
C1	MIDAS-FS-K6	Beta poly, $K_m = 6$	τ_t	constrained	MIDAS-FS
C2	MIDAS-FS-K12	Beta poly, $K_m = 12$	τ_t	constrained	MIDAS-FS
C3	MIDAS-FS-K24	Beta poly, $K_m = 24$	τ_t	constrained	MIDAS-FS
B0	GJR-GARCH	—	—	—	Benchmark

[a] Constrained via sample-mean normalization rather than the $\mathbb{E}[g] = 1$ condition.

MIDAS-RW: rolling-window MIDAS with daily VIX lags. MIDAS-FS: fixed-span MIDAS with monthly VIX averages.

The Beta polynomial $\phi_k(\omega_1, \omega_2) \propto k^{\omega_1 - 1}(K - k)^{\omega_2 - 1}$ with $\omega_1 = 1$ (fixed) and ω_2 estimated.

The key design dimensions are:

1. τ_t **functional form**: VIX-squared ($\tau \propto \text{VIX}^2$) provides dimensional consistency—since VIX is quoted in volatility units, VIX^2 has variance units matching σ_t^2. Log-exponential ($\tau = \exp(\theta_0 + \theta_1 \log \text{VIX})$) is the specification used in the original GARCH-MIDAS literature. VIX-level ($\tau = \exp(\theta_0 + \theta_1 \text{VIX})$) is included for completeness.

2. **Normalization denominator**: Using τ_t (contemporaneous) follows the recursive structure of Engle et al. (2013, Eq. 4), where the current environment determines the normalization. Using τ_{t-1} (lagged) reflects a conservative timing convention in which the normalization denominator is predetermined before the shock r_{t-1} is observed.

3. **Intercept treatment**: The constrained case ($\mathbb{E}[g_t] = 1$) is the standard assumption in component models, ensuring that τ_t alone determines the unconditional variance. The free case provides flexibility to accommodate a systematic wedge between implied and realized variance (the VRP).

3.3 GARCH-MIDAS Benchmark

For the GARCH-MIDAS specifications (B1–B3, C1–C3), the long-run component is:

$$\log \tau_t = m + \theta \sum_{k=1}^{K} \phi_k(\omega_1, \omega_2) \log \mathrm{VIX}_{t-k}, \tag{10}$$

where $\phi_k(\omega_1, \omega_2)$ are Beta-polynomial weights:

$$\phi_k(\omega_1, \omega_2) = \frac{(k/K)^{\omega_1 - 1}(1 - k/K)^{\omega_2 - 1}}{\sum_{j=1}^{K} (j/K)^{\omega_1 - 1}(1 - j/K)^{\omega_2 - 1}}, \tag{11}$$

with $\omega_1 = 1$ (fixed) and ω_2 estimated, following Engle et al. (2013). The rolling-window variant (B1–B3) applies the MIDAS filter at daily frequency with $K \in \{22, 65, 125\}$ daily lags. The fixed-span variant (C1–C3) uses monthly VIX averages with $K_m \in \{6, 12, 24\}$ monthly lags, which is closer to the original GARCH-MIDAS design.

The short-run component g_t follows the same GJR specification as Equation (7), with $\omega_g = 1 - \alpha - \gamma/2 - \beta$.

3.4 Estimation

All multiplicative models are estimated by joint maximum likelihood. For the GARCH-X variants (A-series), the parameter vector is $\boldsymbol{\psi} = (\theta_0, \theta_1, \omega_g, \alpha, \gamma, \beta)$ for the free-omega case, or $\boldsymbol{\psi} = (\theta_0, \theta_1, \alpha, \gamma, \beta)$ with $\omega_g = 1 - \alpha - \gamma/2 - \beta$ for the constrained case. For the GARCH-MIDAS variants (B/C-series), the parameter vector additionally includes m, θ, and ω_2.

Under the assumption of Gaussian innovations, the log-likelihood is:

$$\ell(\boldsymbol{\psi}) = -\frac{1}{2} \sum_{t=1}^{T} \left[\log(2\pi) + \log(\tau_t g_t) + \frac{r_t^2}{\tau_t g_t} \right]. \tag{12}$$

Optimization uses L-BFGS-B with three sets of starting values to mitigate local optima. Parameter bounds ensure positivity ($\alpha, \gamma, \beta > 0$) and stationarity ($\alpha + \gamma/2 + \beta < 1$). For the free-omega models, we additionally impose $\omega_g > 0$ to ensure $g_t > 0$ for all t; in practice, all estimated ω_g values are strictly positive (range 0.012–0.087), so this constraint is never binding.

3.5 Evaluation Framework

3.5.1 Forecast Evaluation

We adopt a rolling-window pseudo-out-of-sample design:

1. Fix an estimation window of $W = 2{,}000$ trading days.
2. Estimate all models on the window $[t - W + 1, t]$.
3. Generate one-step-ahead forecasts $\hat{\sigma}_{t+1}^2$.
4. Advance the window by one day and repeat.
5. Refit parameters every 63 trading days (approximately quarterly) to balance estimation cost and adaptation.

The primary loss function is the QLIKE of Patton (2011):

$$L_{\text{QLIKE}}(\hat{\sigma}_t^2, r_t^2) = \frac{r_t^2}{\hat{\sigma}_t^2} - \log \frac{r_t^2}{\hat{\sigma}_t^2} - 1, \tag{13}$$

evaluated on the squared return r_t^2 as the volatility proxy. The legacy horse-race tables report the average kernel $\log(\hat{\sigma}_t^2) + r_t^2/\hat{\sigma}_t^2$, which differs from Equation 13 by an observation-specific term common to every model in a given comparison. It therefore preserves within-sample loss differences and rankings, but not absolute levels or percentage reductions. Section 6.5 instead reports the full dimensionless loss in Equation 13; levels and percentages must not be compared across those two reporting conventions.[2] The proxy-robustness result of Patton (2011) is an expected-loss result under the required conditional-unbiasedness assumptions for the volatility proxy; it is not a guarantee that a noisy proxy preserves every finite-sample ranking.

Pairwise model comparisons use the Diebold and Mariano (2002) test (hereafter DM test). For the full 16-comparison horse race, we adopt the conservative threshold $|t| > 3.0$, which is close to the Bonferroni-adjusted critical value of $z_{0.05/(2\times16)} \approx 2.95$ and consistent with the multiple-testing concerns raised by White (2000) and Harvey et al. (2016). For the primary A4f vs. GJR comparison, the standard $|t| > 1.96$ threshold applies and is comfortably exceeded.

3.5.2 VaR and ES Backtesting

For risk management evaluation, we compute one-day-ahead Value-at-Risk and Expected Shortfall under both Gaussian and Student-t distributional assumptions:

$$\mathrm{VaR}^{\mathrm{Normal}}_{\alpha,t} = \hat{\sigma}_t\, z_\alpha, \tag{14}$$

$$\mathrm{VaR}^{t}_{\alpha,t} = \hat{\sigma}_t\, t_\alpha(\nu)\sqrt{\frac{\nu-2}{\nu}}, \tag{15}$$

where z_α is the standard normal quantile, $t_\alpha(\nu)$ is the Student-t quantile with ν degrees of freedom, and the scaling factor $\sqrt{(\nu-2)/\nu}$ ensures unit variance of the standardized distribution.

[2] The legacy SPY horse race consequently has values around -8.3, whereas the corrected K1379 diagnostic has values around 1.4–1.5. Each table states its convention and compares models on a common target and sample.

ES is computed as:

$$\mathrm{ES}_{\alpha,t}^{\mathrm{Normal}} = \hat{\sigma}_t \left(-\frac{\phi(z_\alpha)}{\alpha} \right), \tag{16}$$

$$\mathrm{ES}_{\alpha,t}^{t} = \hat{\sigma}_t \left[\frac{f_t(t_\alpha, \nu)(\nu + t_\alpha^2)}{(\nu - 1)\alpha} \right] \sqrt{\frac{\nu - 2}{\nu}}, \tag{17}$$

where $\phi(\cdot)$ is the standard normal density and $f_t(\cdot, \nu)$ is the Student-t density.

Backtesting employs four tests:

1. **Kupiec (1995) unconditional coverage** (UC): tests whether the violation rate equals the nominal level.

2. **Christoffersen (1998) conditional coverage** (CC): additionally tests independence of violations.

3. **Engle & Manganelli (2004) dynamic quantile** (DQ): tests whether violations are predictable.

4. **Acerbi & Szekely (2019)** Z1/Z2 tests for ES adequacy.

A model "passes" at level α if all three VaR tests (UC, CC, DQ) have $p > 0.05$.

4 Data

4.1 Primary Sample

Our main analysis uses daily closing prices of the SPDR S&P 500 ETF Trust (SPY) and the CBOE Volatility Index (VIX) from January 3, 2005, to April 7, 2026, obtained from Yahoo Finance. After computing log returns and aligning the VIX series, the sample contains $T = 5{,}347$ trading days.

The out-of-sample period begins on January 2, 2019, providing $n_{\mathrm{OOS}} = 1{,}825$ daily observations. This period encompasses the COVID-19 crash of March 2020 (VIX peak of 82.69), the post-pandemic recovery, the 2022 inflation-driven drawdown, and the 2025 trade-war volatility spike, providing a demanding test environment that includes multiple volatility regimes.

Table 2 reports summary statistics.

Table 2: Summary statistics for SPY daily log returns and VIX

	Full Sample		OOS (2019–2026)	
Statistic	SPY r_t	VIX	SPY r_t	VIX
Mean (ann.)	—	18.97	—	22.14
Std (ann.)	0.188	—	0.213	—
Skewness	−0.37	2.15	−0.82	1.89
Kurtosis	14.52	9.87	12.31	7.03
Min	−0.128	9.14	−0.128	12.10
Max	0.116	82.69	0.093	82.69
N	5,347	5,347	1,825	1,825

SPY log returns and VIX closing levels. Annualized mean and standard deviation for returns. Kurtosis is excess kurtosis. Data source: Yahoo Finance.

4.2 Cross-Asset Sample

For cross-asset validation (Section 5.3), we additionally use:

- **QQQ** (Invesco QQQ Trust): high-beta U.S. technology exposure, high VIX correlation.
- **EEM** (iShares MSCI Emerging Markets ETF): moderate VIX correlation.
- **GLD** (SPDR Gold Shares): low VIX correlation. We additionally use the CBOE Gold ETF Volatility Index (GVZ) as an asset-specific fear index.
- **0050.TW** (Yuanta/P-shares Taiwan Top 50 ETF): Taiwan equity market with VIX lag of one additional day due to time-zone differences.

All data are obtained from Yahoo Finance for the same sample period. The 0050.TW data are cleaned using the split-adjustment procedure of Lai (2026) to correct for the unadjusted 1:4 stock split prior to 2014.

5 Empirical Results

5.1 Specification Horse Race

Table 3 presents the complete out-of-sample results for all 17 specifications, ranked by QLIKE loss.

Table 3: Out-of-sample specification comparison: QLIKE rankings

Rank	Model	QLIKE	DM t vs GJR	Harvey sig.	Category
1	**A4f** (VIX^2, free ω)	**−8.360**	**4.03**	**Yes**	GARCH-X
2	A4 (VIX^2, constr.)	−8.358	3.81	Yes	GARCH-X
3	A2 (log-exp, τ_t)	−8.355	3.24	Yes	GARCH-X
4	A2f (log-exp, free ω)	−8.354	3.22	Yes	GARCH-X
5	B1 (MIDAS-RW, $K = 22$)	−8.354	3.21	Yes	MIDAS-RW
6	A3 (log-exp, τ_{t-1})	−8.350	3.05	Yes	GARCH-X
7	A2n (log-exp, norm)	−8.350	3.07	Yes	GARCH-X
8	A4n (VIX^2, norm)	−8.350	3.45	Yes	GARCH-X
9	A1 (original)	−8.349	2.99	No	GARCH-X
10	A3f (τ_{t-1}, free ω)	−8.347	2.92	No	GARCH-X
11	B2 (MIDAS-RW, $K = 65$)	−8.342	2.99	No	MIDAS-RW
12	B3 (MIDAS-RW, $K = 125$)	−8.337	3.06	Yes	MIDAS-RW
13	A5 (VIX level)	−8.332	1.90	No	GARCH-X
14	C1 (MIDAS-FS, $K_m = 6$)	−8.310	3.49	Yes	MIDAS-FS
15	C3 (MIDAS-FS, $K_m = 24$)	−8.307	2.68	No	MIDAS-FS
16	C2 (MIDAS-FS, $K_m = 12$)	−8.301	2.13	No	MIDAS-FS
17	B0 (GJR-GARCH)	−8.273	ref	—	Benchmark

QLIKE evaluated on r_t^2 following Patton (2011). DM t is the Diebold-Mariano test statistic against GJR-GARCH (B0); positive values indicate the model is better. Harvey significance requires $|t| > 3.0$. OOS period: 2019-01-01 to 2026-04-07, n = 1,825. Estimation window W = 2,000, refit every 63 days.

Several findings emerge from Table 3.

Finding 1: VIX-squared is the best functional form for τ_t. The top two models both use $\tau_t = \theta_0 + \theta_1 \mathrm{VIX}_{t-1}^2$. This dominance has a simple explanation: VIX is quoted in annualized volatility units (percent), so VIX^2 has annualized variance units, directly matching the dimension of $\sigma_t^2 = \tau_t \times g_t$. The log-exponential specification (A2, A3) performs well but introduces a nonlinear transformation that is not dimensionally motivated. The VIX-level specification (A5) performs worst among the GARCH-X models, failing to reach the Harvey threshold ($t = 1.90$).

Finding 2: Contemporaneous normalization (τ_t) dominates lagged (τ_{t-1}). Comparing A2 vs. A3 (both log-exp, constrained), A2 achieves a lower QLIKE (-8.355 vs. -8.350). This ordering is consistent with the logic of Engle et al. (2013, Eq. 4), where $u_{t-1} = r_{t-1}/\sqrt{\tau_t}$ uses the *current* environment to normalize past shocks.

Finding 3: Free omega improves the VIX-squared model. Comparing A4f vs. A4, freeing ω_g reduces QLIKE from -8.358 to -8.360 and increases the DM t from 3.81 to 4.03. This improvement is consistent with the presence of a positive variance risk premium: VIX^2 systematically exceeds realized variance, so constraining $\mathbb{E}[g_t] = 1$ forces an inappropriate normalization. However, free omega does *not* improve the log-exp model (A2f vs. A2: -8.354 vs. -8.355), suggesting that the log-exp parameterization already absorbs some of this flexibility through its intercept θ_0.

Finding 4: The best daily GARCH-X point estimate outperforms GARCH-MIDAS, but the category rankings overlap. A4f ranks ahead of every GARCH-MIDAS variant in Table 3, and several other daily specifications rank ahead of the best MIDAS model, B1. The weaker daily VIX-level specification A5, however, ranks below B1–B3, so this is not category-wide dominance. The Model Confidence Set analysis (Section 6) likewise shows that A4f is indistinguishable from B1 ($K = 22$) in pairwise DM tests ($t = -0.90$). The point-estimate advantage of A4f is consistent with VIX information being concentrated in the most recent observation, which the MIDAS Beta-polynomial weighting dilutes across 22–125 lags, but parsimony rather than statistical dominance is the practical argument for preferring the daily specification.

Finding 5: The A1 specification reveals the cost of inconsistency. The original A1 specification, which uses τ_t in estimation but τ_{t-1} in OOS forecasting, ranks 10th. Adopting a consistent denominator (either τ_t everywhere as in A2, or τ_{t-1} everywhere as in A3) improves performance, confirming that estimation/forecasting consistency is important in multiplicative models.

5.2 Source Decomposition and the Variance Risk Premium

The multiplicative structure $\sigma_t^2 = \tau_t \times g_t$ admits an economic interpretation when $\tau_t \approx \theta_0 + \theta_1 \mathrm{VIX}_{t-1}^2$. In this case:

$$g_t = \frac{\sigma_t^2}{\tau_t} \approx \frac{\text{realized variance}}{\text{function of implied variance}}. \tag{18}$$

When $\mathbb{E}[g_t] = 1$ (constrained case), τ_t fully determines the unconditional variance, and g_t fluctuates around unity, rising when realized variance exceeds what VIX implies (low VRP or variance sellers losing) and falling when VIX over-estimates realized variance (high VRP or variance sellers profiting). When $\mathbb{E}[g_t] \neq 1$ (free case), the model accommodates the average VRP through ω_g.

We label this a *source decomposition* rather than the traditional *long-run/short-run decomposition* used in the GARCH-MIDAS literature. In our framework:

- τ_t = *exogenous source*: the aggregate fear level determined by option market participants, incorporating forward-looking information from across the options surface.
- g_t = *endogenous source*: the GARCH dynamics that capture how rapidly realized volatility responds to—and deviates from—the options-market consensus.

This interpretation differs from "long-run vs. short-run" because τ_t updates daily (it is not a slow-moving trend) and because both components have comparable persistence.

5.2.1 VRP Tracking

To quantify the connection between g_t and VRP, we define the VRP proxy as:

$$\mathrm{VRP}_t = \frac{\mathrm{VIX}_{t-1}^2}{252} - r_t^2, \tag{19}$$

following Bollerslev et al. (2009), where $\mathrm{VIX}^2/252$ converts the annualized implied variance to a daily scale.

Table 4 reports the Spearman rank correlation between the model-implied g_t and the VRP proxy, computed over the OOS period.

Table 4: Spearman correlation between model g_t component and VRP proxy

Measure	Spearman ρ	p-value
g_t (A3f: τ_{t-1}, free ω)	0.819	< 0.001
g_t (A2n: log-exp, normalized)	0.801	< 0.001
g_t (A4n: VIX^2, normalized)	0.778	< 0.001
g_t (A4f: VIX^2, free ω, *recommended*)	-0.017	0.479
Raw r_t^2/VIX_{t-1}^2	0.151	< 0.001

VRP proxy: $\text{VIX}_{t-1}^2/252 - r_t^2$. OOS period: 2019–2026, $n = 1{,}824$. Spearman ρ used for robustness to outliers.

A4f's near-zero $\rho = -0.017$ ($p = 0.479$) reflects that free ω_g absorbs the average VRP level (so $\mathbb{E}[g_t] = 0.48$ rather than 1), leaving g_t to capture residual GARCH dynamics approximately orthogonal to VRP. The high correlations for A3f/A2n/A4n arise because their τ_t specifications absorb less VRP signal, leaving a larger systematic component in g_t.

The GARCH-based g_t achieves $\rho = 0.78$–0.82 with VRP, vastly exceeding the raw ratio ($\rho = 0.15$). We acknowledge that part of this correlation is *mechanical*: by construction, $g_t \approx \sigma_t^2/\tau_t$, which is related to VRP through the variance decomposition. However, the dramatic improvement from $\rho = 0.15$ (raw ratio) to $\rho \approx 0.80$ (GARCH-filtered) is *not* mechanical—it arises because the GARCH autoregressive structure in g_t smooths the highly noisy daily r_t^2/VIX^2 ratio, extracting its persistent component. The GARCH dynamics are the empirical contribution; the ratio structure is the mechanical starting point. This demonstrates that the multiplicative model does not merely scale variance by VIX—it provides a disciplined framework for decomposing realized variance into option-market expectations and residual dynamics.

5.2.2 VRP Predictability: A Null Result

Given the strong contemporaneous relationship between g_t and VRP, a natural question is whether g_t can *predict* future VRP. We test this using four approaches:

1. **Granger causality**: g_t Granger-causes VRP at horizons $h = 1, 5, 10$ days ($p < 0.001$) but not at $h = 22$ days ($p = 0.18$).

2. **Predictive regressions**: In specifications controlling for lagged VRP, the Newey and West (1987) t-statistic on g_t ranges from -1.28 to -1.95. Augmenting the regression with additional controls (log-VIX, lagged returns) increases the t-statistic to -2.34 at the $h = 5$ horizon, but no specification surpasses the Harvey $|t| > 3.0$ threshold.

3. **OOS R^2**: Following Campbell and Thompson (2008), the OOS R^2 of g_t-based forecasts relative to the historical mean is 1.3% at $h = 1$ and negative at longer horizons, indicating overfitting.

4. **Variance swap strategy**: A strategy that sells variance when $g_t > 1$ ("cheap variance") and buys when $g_t < 1$ produces a Sharpe ratio of -0.64, compared to $+0.85$ for the naive always-sell strategy.

We report this null result as an honest finding: g_t **is a contemporaneous VRP proxy, not a VRP predictor.** This is consistent with the low daily VRP autocorrelation ($\rho \approx 0.20$) and the well-known difficulty of predicting variance risk premia at short horizons (Bollerslev et al., 2009).

5.3 Cross-Asset Validation

Table 5 reports the performance of A4f across six additional assets.

Table 5: Cross-asset validation of A4f specification

Asset	VIX-r^2 corr.	GJR QLIKE	A4f QLIKE	DM t	Harvey sig.
SPY (primary)	0.52	-8.277	-8.361	4.03	**Yes**
QQQ	0.49	1.481	1.420	3.71	**Yes**
GLD (with GVZ)	0.38^a	1.530	1.457	3.17	**Yes**
STOXX50E	0.44	1.565	1.513	3.64	**Yes**
FEZ	0.44	1.422	1.371	3.45	**Yes**
EEM	0.50	1.338	1.307	2.47	No
0050.TW	0.28	1.458	1.432	1.44	No

DM t: positive values indicate A4f has lower (better) QLIKE than GJR. Harvey significance: $|t| > 3.0$. OOS: 2019–2026. For GLD, A4f uses GVZ: $\tau_t = \theta_0 + \theta_1 \mathrm{GVZ}_{t-1}^2$. STOXX50E and FEZ use US VIX (not VSTOXX), which suffices due to global fear synchronization.

The SPY row reports the legacy QLIKE kernel on decimal returns; the six additional-asset rows report full dimensionless QLIKE. Loss levels are therefore comparable only within an asset, while each DM test compares GJR and A4f on the same asset, target, and convention.

[a] GVZ-r^2 correlation for GLD.

Multiple-testing note: For the 6 validation assets beyond the primary SPY result, the two-sided 5% Bonferroni-adjusted normal threshold is $|t| > z_{1-0.05/(2\times6)} \approx 2.64$. QQQ ($t = 3.71$), STOXX50E ($t = 3.64$), FEZ ($t = 3.45$), and GLD with GVZ ($t = 3.17$) remain above this validation-family threshold; EEM and 0050.TW do not. Primary SPY ($t = 4.03$) is reported separately and is not counted among those six validation tests.

The A4f specification achieves Harvey-significant improvements for five of seven assets: SPY ($t = 4.03$), QQQ ($t = 3.71$), GLD with GVZ ($t = 3.17$), EURO STOXX 50 ($t = 3.64$), and FEZ ($t = 3.45$). Notably, the European results use US VIX rather than VSTOXX, suggesting that VIX captures a *global* fear factor that extends beyond U.S. equities. The model fails to reach significance for EEM ($t = 2.47$) and 0050.TW ($t = 1.44$), likely because these markets have stronger idiosyncratic drivers.

The pattern is informative: the model works when the fear index is *relevant* to the asset. SPY and QQQ are both U.S. equity products for which VIX is the natural fear gauge. GLD benefits from its own implied volatility index (GVZ). EEM and 0050.TW are driven by idiosyncratic factors that U.S. VIX does not capture—consistent with our source decomposition interpretation, which requires τ_t to reflect the asset's *own* fear environment.

Table 6 examines whether asset-specific fear indices can improve the model for EEM and 0050.TW.

Table 6: Local fear index experiments for EEM, GLD, and 0050.TW

Asset	Fear Index	QLIKE	DM t vs GJR	Harvey sig.
GLD	VIX (U.S. equity)	1.513	1.08	No
	GVZ (gold vol)	1.457	3.17	**Yes**
	VIX + GVZ (dual)	1.457	3.39	**Yes**
EEM	VIX	1.306	2.52	No
	OwnRV_{20}	1.339	-0.60	No
	VIX + OwnRV_{20}	1.306	2.60	No
0050.TW	VIX (lag+1)	1.431	1.68	No
	OwnRV_{20}	1.460	-0.16	No
	VIX + OwnRV_{20}	1.430	1.70	No
	VIX (lag+2)	1.440	1.55	No

OwnRV_{20}: 20-day realized volatility of the asset itself. Dual factor: $\tau_t = \theta_0 + \theta_1 X_1^2 + \theta_2 X_2^2$. OOS: 2019–2026.

The GLD result is particularly instructive: switching from U.S. VIX ($t = 1.08$, not significant) to the asset-specific GVZ ($t = 3.17$, significant) dramatically improves performance, confirming that the multiplicative structure requires a *relevant* implied volatility measure. For EEM and 0050.TW, neither the own-asset realized volatility nor dual-factor models achieve significance, likely because no high-quality implied volatility index exists for these markets.

5.4 VaR and Expected Shortfall Backtesting

Table 7 reports the VaR backtesting results for SPY at three confidence levels ($\alpha = 1\%, 2.5\%, 5\%$) under four model-distribution combinations.

Table 7: VaR backtesting results for SPY (OOS 2019–2026)

Model	α	Viol.%	Expected	UC p	CC p	DQ p	Pass?
GJR-Normal	1%	2.30	1.00	0.000	0.000	0.000	Fail
GJR-Normal	2.5%	3.78	2.50	0.001	0.004	0.029	Fail
GJR-Normal	5%	6.19	5.00	0.024	0.073	0.074	Fail
GJR-t	1%	1.64	1.00	0.011	0.033	0.017	Fail
GJR-t	2.5%	3.45	2.50	0.014	0.048	0.064	Fail
GJR-t	5%	6.30	5.00	0.014	0.039	0.023	Fail
A4f-Normal	1%	1.92	1.00	0.000	0.002	0.000	Fail
A4f-Normal	2.5%	3.29	2.50	0.040	0.088	0.005	Fail
A4f-Normal	**5%**	**5.15**	5.00	**0.769**	**0.830**	**0.433**	**Pass**
A4f-t	**1%**	**1.42**	1.00	**0.087**	**0.158**	**0.354**	**Pass**
A4f-t	2.5%	3.01	2.50	0.173	0.337	0.006	Fail
A4f-t	**5%**	**5.42**	5.00	**0.411**	**0.554**	**0.172**	**Pass**

"Pass" requires all three tests (UC, CC, DQ) to have $p > 0.05$. Viol.% is the empirical violation rate. UC: Kupiec (1995). CC: Christoffersen (1998). DQ: Engle and Manganelli (2004). GJR-t uses jointly estimated degrees of freedom (mean $\hat{\nu} = 5.94$). A4f-t estimates ν from standardized residuals (mean $\hat{\nu} = 8.12$). $n_{\mathrm{OOS}} = 1{,}825$.

The results are striking: GJR-GARCH fails all VaR tests at all confidence levels, regardless of distributional assumption. A4f-Normal passes only at the 5% level, but A4f with Student-t innovations passes at both the 1% and 5% levels. The improvement in violation rates is substantial: at the 1% level, A4f-t achieves 1.42% (close to the nominal 1%) versus GJR-Normal's 2.30% (more than double the nominal rate).

Table 8 reports the ES backtesting results at $\alpha = 2.5\%$.

Table 8: Expected Shortfall backtesting at $\alpha = 2.5\%$ (SPY, OOS 2019–2026)

Model	Z1 stat	Z1 p	Z2 stat	Z2 p	Pass?
GJR-Normal	2.200	0.502	2.816	0.539	Pass
GJR-t	2.098	0.502	2.516	0.537	Pass
A4f-Normal	2.119	0.516	2.471	0.524	Pass
A4f-t	2.058	0.510	2.275	0.526	Pass

ES tests following Acerbi and Szekely (2019). Z1 tests ES level adequacy; Z2 tests ES conditional calibration. All models pass both tests.

All models pass the ES tests, which is consistent with the known lower power of ES backtests relative to VaR backtests. The combined VaR+ES scorecard (Table 9) summarizes the risk management performance.

Table 9: Combined VaR/ES scorecard

Model	VaR Pass (of 3)	ES Pass (of 1)	Total (of 4)
GJR-Normal	0	1	1
GJR-t	0	1	1
A4f-Normal	1	1	2
A4f-t	**2**	**1**	**3**

The results demonstrate that A4f provides not only superior point forecasts (QLIKE) but also improved tail risk measurement. The combination of multiplicative structure (better mean prediction) and Student-t innovations (better tail modeling) yields a risk management model that passes the majority of backtests—a practically relevant finding for risk officers and regulators.

6 Robustness

6.1 Sensitivity to Estimation Settings

We conduct a comprehensive sensitivity analysis across four design dimensions: refit frequency, estimation window size, OOS sub-period, and alternative VIX indices. Table 10 reports the results.

Table 10: Sensitivity analysis of A4f specification

Dimension	Setting	QLIKE (GJR)	QLIKE (A4f)	DM t	Harvey sig.
Refit freq.	21 days	1.500	1.409	4.29	Yes
	63 days (baseline)	1.498	1.408	3.92	Yes
	126 days	1.501	1.408	3.36	Yes
	252 days	1.509	1.410	3.32	Yes
Window	$W = 1{,}000$	1.478	1.418	3.18	Yes
	$W = 1{,}500$	1.490	1.408	3.49	Yes
	$W = 2{,}000$ (baseline)	1.498	1.408	3.92	Yes
	$W = 2{,}500$	1.491	1.402	5.13	Yes
	$W = 3{,}000$	1.487	1.402	4.94	Yes
Sub-period	2019–2020 (COVID)	1.522	1.408	1.60	No[a]
	2021–2022 (rate hikes)	1.384	1.303	2.50	No[a]
	2023–2026 (stable)	1.552	1.473	4.52	Yes
VIX variant	VIX (baseline)	1.498	1.408	3.92	Yes
	VIX9D	1.498	1.380	5.15	Yes
	VIX3M	1.498	1.436	2.59	No
	VIX/VIX3M ratio	1.498	1.446	3.53	Yes

[a] Sub-period DM tests have reduced power ($n \approx 500$); QLIKE improvement is directionally consistent in all sub-periods.

VIX9D data begins 2011; sub-period and VIX variant tests use the shorter sample. SPY data from Yahoo Finance.

Results replicated in a sensitivity rerun (2026-04-17); DM t values differ by $|\Delta| \leq 0.82$ across cells (median $|\Delta| = 0.27$) due to a 5-day data vintage difference; Harvey significance count unchanged at 13/16. Replication script and cell-by-cell diff available in `experiments/k988_sens/`.

A4f passes the Harvey threshold in 13 of 16 sensitivity settings (81.3%). All four refit frequencies and all five window sizes produce significant results, demonstrating that the model is not sensitive to estimation design choices. Notably, longer estimation windows ($W = 2{,}500$–$3{,}000$) strengthen the DM statistic (from 3.92 to 5.13), suggesting that A4f benefits from larger training samples.

The sub-period results show that A4f improves QLIKE in all three sub-periods, but the DM test lacks power with $n \approx 500$ observations in the COVID and rate-hike periods. This is a well-known limitation of the DM test in short samples, not evidence of model failure.

To address the concern that A4f's advantage may be driven by a single extreme episode (e.g., the COVID-19 crash), we conduct an extended sub-period analysis using seven non-overlapping two-year windows spanning 2013–2026. A4f achieves lower QLIKE than GJR-GARCH in all seven periods (7/7), with improvements ranging from 4.80% to 7.00% (mean 6.42%). The COVID period (2019–2020) shows the largest improvement (7.00%), but removing it leaves the remaining six periods with improvements of 4.80–6.90%—confirming that the model's advantage is not COVID-dependent. Of the seven individual periods, three pass the Harvey $|t| > 3.0$ threshold and six pass $|t| > 2.0$; the pooled full-period DM statistic is $t = 6.977$, far exceeding conventional thresholds. These results demonstrate that A4f's superiority is robust across low-volatility environments (2017–2018), crisis periods (2019–2020, 2025–2026), and recovery phases (2021–2022, 2023–2024).

A complementary leave-COVID-out test provides a more stringent robustness check. Rather than splitting the OOS window into two-year periods, we remove the 104 trading days spanning 2020-02-01 to 2020-06-30 from the full OOS window and re-estimate the DM test on the remaining $n = 1{,}721$ observations. Table 11 reports the results, computed in a dedicated baseline-faithful replication run (`experiments/k1393/`); the comparison full-OOS benchmark $t = 4.03$ comes from the canonical pairwise DM matrix in `mcs_dm_results.json` used throughout Table 3, so the two statistics share specification but differ in OOS loss-path source. The non-COVID DM statistic ($t = 4.26$) *exceeds* the primary full-OOS result (Table 3: $t = 4.03$), comfortably surpassing the Harvey $|t| > 3.0$ threshold—the opposite of what would be expected if the COVID-era volatility spike were the primary driver. Post-COVID observations (2021–2026, $n = 1{,}448$) independently confirm Harvey significance ($t = 3.76$). Within the COVID window itself

(n = 104), the DM statistic drops to t = 1.48, not because A4f underperforms—the mean QLIKE improvement during this window is approximately 2.5× larger than outside it—but because 104 observations are insufficient for the HAC-corrected DM test to reach conventional significance. These results confirm that A4f's advantage is a persistent normal-market phenomenon, not an artifact of a single extreme episode.

Table 11: Leave-COVID-out robustness: DM tests for A4f vs. GJR-GARCH by OOS sub-sample

Sub-sample (OOS: 2019-01-01–2026-04-07)	n	DM t	Harvey sig. ($\lvert t\rvert > 3.0$)	Interpretation
Non-COVID (excl. 2020-02-01–2020-06-30)	**1,721**	**4.26**	**Yes**	Advantage strengthen
Pre-COVID (2019)	273	2.52	No[a]	Short-sample
COVID window (2020-02–06)	104	1.48	No[a]	Extreme short-sampl
Post-COVID (2021–2026-04)	1,448	3.76	Yes	Confirmed

A4f specification: $\tau_t = \theta_0 + \theta_1 \mathrm{VIX}^2_{t-1}$, free ω_g, Gaussian quasi-likelihood; OOS window W = 2,000, refit every 63 day DM test uses HAC-robust standard errors (Newey–West with bandwidth $q = \lfloor n^{1/3} \rfloor$, which yields $q \in \{4, 6, 11, 11, 12$ for the COVID-window, pre-COVID, non-COVID, post-COVID, and full-OOS sub-samples respectively). For th primary full-OOS DM result (t = 4.03, n = 1,825), see Table 3.

[a] Insufficient power at $n < 300$; mean QLIKE improvement is directionally consistent across all sub-samples.

Source: `experiments/k1393/k1393_results.json`.

Among VIX variants, the 9-day VIX (VIX9D) achieves the strongest result (DM t = 5.15), consistent with short-term implied volatility being more responsive to recent market conditions. The 3-month VIX (VIX3M) does not reach significance (t = 2.59), suggesting that longer-term implied volatility changes too slowly to provide daily forecasting value.

6.2 Estimation/Forecasting Consistency

The comparison between A1 (inconsistent denominator) and A2/A3 (consistent) demonstrates the importance of using the same τ normalization in estimation and forecasting. A1 ranks 10th, while A2 ranks 3rd and A3 ranks 7th, confirming that consistency matters for out-of-sample performance. This is a methodological contribution relevant to implementing multiplicative models in practice.

6.3 Model Confidence Set

To assess whether A4f can be statistically distinguished from competing specifications, we apply the Model Confidence Set (MCS) procedure of Hansen et al. (2011) with 1,000 bootstrap replications. At $\alpha = 0.10$, all 17 models survive—the pairwise QLIKE differences among the VIX-augmented models are too small to reject any specification. At $\alpha = 0.25$, the MCS eliminates only GJR-GARCH ($p = 0.229$); all 16 VIX-augmented models remain.

This result qualifies our ranking-based conclusions: while A4f achieves the lowest QLIKE, it is not *statistically* distinguishable from other well-specified GARCH-X or GARCH-MIDAS models incorporating VIX. A4f and a majority of the VIX-augmented specifications exceed the $|t| > 3$ pairwise screen against GJR-GARCH, but the weaker specifications do not; the MCS result is a set-level statement, not evidence that every VIX model wins pairwise. A4f is the point-estimate best among the alternatives considered.

6.4 Pairwise DM Tests Among Top Models

Table 12 reports pairwise DM tests between A4f and selected competitors to clarify which differences are statistically significant.

Table 12: Pairwise DM tests: A4f vs. selected competitors

Comparison	DM t	Harvey sig.	Interpretation
A4f vs. GJR (B0)	-4.03	Yes	A4f dominates benchmark
A4f vs. A4 (constr.)	-0.64	No	Free ω adds little
A4f vs. A2 (log-exp)	-0.72	No	Functional form indistinguishable
A4f vs. B1 (MIDAS-RW-22)	-0.90	No	Best MIDAS $\approx$ A4f
A4f vs. B2 (MIDAS-RW-65)	-3.04	Yes	A4f $>$ longer-lag MIDAS
A4f vs. B3 (MIDAS-RW-125)	-3.73	Yes	A4f $>$ longest-lag MIDAS
A4f vs. C1 (MIDAS-FS-6)	-2.45	No	Borderline

The loss differential is first model minus second model, so negative t favors the first model. DM test with Harvey et al. (2016) threshold $|t| > 3.0$.

A4f significantly outperforms GJR-GARCH and the longer-lag MIDAS specifications (B2, B3), but is statistically indistinguishable from the best-performing GARCH-X alternatives (A4, A2) and the shortest-lag MIDAS (B1, $K = 22$). This confirms the MCS finding: the key distinction is VIX-augmented vs. plain GARCH, not the specific functional form of the VIX component.

The Giacomini and White (2006) conditional predictive ability test further supports this interpretation: A4f conditionally outperforms GJR ($\chi^2 = 16.28$, $p < 0.001$) but not B1 ($\chi^2 = 3.77$, $p = 0.152$).

6.5 HAR-Style Daily-r^2 Benchmark Comparison

A canonical HAR-RV benchmark uses intraday realized variance (Corsi, 2009), which is not available in the pinned daily-data packet used here. We therefore report a narrower diagnostic: HAR-style regressions whose daily, weekly, and monthly components are formed from lagged squared daily returns, with and without VIX^2_{t-1}. These models are not canonical HAR-RV specifications and cannot close the realized-measure benchmark question by themselves. All models are evaluated against the same r_t^2 proxy using Patton QLIKE, $L_t = r_t^2/\hat{\sigma}_t^2 - \log(r_t^2/\hat{\sigma}_t^2) - 1$.

The corrected K1379 run uses a hash-pinned unique-date SPY/VIX snapshot, fixes the original endpoint at 18 May 2026, and applies the same rolling protocol ($W = 2{,}000$, refit every 63 days). There are 1,852 shared valid OOS losses. A4f has mean QLIKE 1.400, compared with 1.480 for GJR and 1.524 for the HAR-style daily-r^2 model, reductions of 5.4% and 8.2%, respectively. Here the loss differential is $d_t = L_{\text{A4f},t} - L_{\text{comparison},t}$, so negative t favors A4f. This is the opposite of Table 3, whose against-GJR column reports GJR loss minus model loss. Bartlett-HAC DM tests with the canonical rule-selected maximum lag of 13 give:

- A4f vs. GJR: DM $t = -4.37$, $p = 0.000013$;
- A4f vs. HAR-style daily-r^2: DM $t = -7.70$, $p < 10^{-13}$.

Both comparisons exceed the $|t| > 3$ reporting screen of Harvey et al. (2016). The two

A4f comparisons also remain beyond that screen over HAC lags 0, 1, 5, 10, 13, and 20. By contrast, the HAR-style daily-r^2 minus GJR comparison is bandwidth-sensitive: its positive statistic favors GJR, and its canonical lag-13 value is $t = 3.11$, versus 2.94 and 2.97 at lags 0 and 1.

The VIX-augmented HAR-style OLS specification is numerically unstable: it produces three nonpositive raw variance forecasts, which are bounded at 10^{-16} before QLIKE and make its full-sample mean loss floor-dependent. Its A4f comparison does not cross the Harvey screen either in the bounded primary run ($t = -1.05$, $p = 0.294$) or in the diagnostic common sample excluding those three dates ($t = -2.26$, $p = 0.024$). We therefore do not interpret its extreme full-sample mean loss as an economic effect.

These results supersede the earlier draft statistics ($t = +0.29$ and $+0.65$), which were computed with the QLIKE ratio reversed, without an HAC long-run variance, and with inconsistent A4f normalization between fitting and OOS recursion. The former approximation-based comparison was not an independent proxy check: it also used daily r^2 components and a different steady-state approximation for A4f. Accordingly, we withdraw the claims of cross-proxy consistency, parity, and statistical non-inferiority. The evidence supports lower loss for A4f relative to the stable daily-r^2 diagnostic under this fixed-window protocol; it does not establish equivalence or performance relative to canonical intraday HAR-RV, so the broader benchmark limitation remains open.

6.6 Multiple Testing Considerations

With 16 pairwise tests against GJR, 10 of 16 multiplicative models achieve $|t| > 3.0$ at the Harvey threshold. Our main result—A4f vs. GJR with $t = 4.03$—exceeds the Bonferroni-adjusted critical value of $z_{0.05/(2\times16)} \approx 2.95$.

6.7 Student-t Degrees of Freedom

The A4f-t model estimates degrees of freedom from standardized residuals $\hat{\varepsilon}_t = r_t/\hat{\sigma}_t$ in each estimation window, yielding mean $\hat{\nu} = 8.12$ (range 7.24–9.84). This is higher than the jointly estimated $\hat{\nu} = 5.94$ (range 5.46–7.67) for GJR-t, consistent with the VIX component absorbing some tail behavior that GJR captures solely through fat-tailed innovations.

6.8 Residual Diagnostics: How VIX Absorption Reduces Tail Risk

The shift in optimal degrees of freedom documented above motivates a deeper question: *how does the VIX component change the distributional properties of standardized residuals?* If A4f's τ_t absorbs the extreme volatility episodes that VIX captures, the standardized residuals $\hat{\varepsilon}_t = r_t/\hat{\sigma}_t$ should exhibit lighter tails than those from GJR-GARCH.

Table 13 reports a comprehensive diagnostic comparison of standardized residuals from GJR-t and A4f-t over the OOS period (n = 1,828).

Table 13: Residual diagnostics: GJR-t vs. A4f-t standardized residuals (OOS)

Diagnostic	GJR-t	A4f-t	Change	Interpretation
Excess kurtosis	3.065	1.238	−59.6%	Substantially lighter tails
Skewness	−0.856	−0.594	−30.6%	Reduced left-tail asymmetry
Jarque-Bera statistic	938.8	224.2	−76.1%	Closer to normality
Median Student-t $\hat{\nu}$	5.28	8.00	+51.5%	Shift toward Gaussian

Standardized residuals $\hat{\varepsilon}_t = r_t/\hat{\sigma}_t$. Excess kurtosis relative to normal (= 0). Student-t $\hat{\nu}$ estimated by MLE in each rolling window; median across all OOS windows reported. SPY, OOS 2019–2026, n = 1,828.

The results are striking: A4f reduces excess kurtosis by 60% (from 3.065 to 1.238), the Jarque-Bera statistic by 76% (from 938.8 to 224.2), and shifts the optimal Student-t degrees of freedom from 5.28 to 8.00—moving the residual distribution substantially toward the Gaussian limit. The skewness reduction (−30.6%) indicates that A4f also absorbs part of the left-tail asymmetry associated with market crashes.

This finding has a direct economic interpretation: extreme VIX spikes (such as the COVID-19 peak of VIX = 82.69) generate heavy-tailed returns that GJR-GARCH must accommodate entirely through fat-tailed innovations (low ν). By absorbing these events through the τ_t component, A4f "normalizes" the residuals, allowing the distributional assumption to focus on the *remaining* non-Gaussianity rather than compensating for unmodeled volatility regime shifts.

This mechanism directly explains the VaR improvement documented in Section 5.4: better-calibrated residual distributions lead to more accurate quantile estimates, particularly in the tails where kurtosis matters most.

6.9 VIX vs. Macroeconomic Variables

A natural question is whether macroeconomic variables can substitute for or complement VIX in the multiplicative framework. Following Conrad and Loch (2015), we estimate GARCH-X models with six macroeconomic variables (term spread, unemployment rate, industrial production, consumer sentiment, financial stress index, and inflation expectations) as alternatives to VIX. The market-implied VIX dominates all macroeconomic specifications (DM $t = 4.77$ for VIX vs. best macro model), consistent with VIX aggregating all publicly available volatility-relevant information through option prices.

7 Discussion: Theoretical Foundation of Source Decomposition

A natural concern is whether the "source decomposition" label is merely a relabeling of the standard long-run/short-run narrative in GARCH-MIDAS models. We address this by deriving two propositions and one empirical remark that establish the economic content of the decomposition.

Proposition 1 (Unconditional Variance Identity)**.** *Under the constrained model ($\omega_g = 1 - \alpha - \gamma/2 - \beta$), $\mathbb{E}[g_t] = 1$ in the stationary distribution, and the unconditional variance satisfies:*

$$\mathbb{E}[\sigma_t^2] = \mathbb{E}[\tau_t] \cdot \mathbb{E}[g_t] + \mathrm{Cov}(\tau_t, g_t). \tag{20}$$

Note: This is an algebraic identity that follows directly from the multiplicative structure;

it holds for any stationary τ_t and g_t and does not require estimation.

This identity is exact, not an approximation. The simpler $\mathbb{E}[\sigma_t^2] \approx \mathbb{E}[\tau_t]$ holds only when $\text{Cov}(\tau_t, g_t) \approx 0$. Empirically, $\text{Corr}(\tau_t, g_t) \approx 0.49$, reflecting the crisis channel: high VIX $\to$ high τ_t $\to$ large standardized shocks $\to$ higher g_t.

Proposition 2 (VRP Auto-Correction)**.** *Under the constrained model (*$\mathbb{E}[g_t] = 1$*, with* $\omega_g = 1 - \alpha - \gamma/2 - \beta$*), when* $\tau_t = \theta_0 + \theta_1 \text{VIX}_{t-1}^2$*, the MLE of* θ_1 *endogenously corrects for the variance risk premium. Specifically, the ratio* $\theta_1/[1/(252 \times 10{,}000)]$ *measures the discount on implied variance relative to the "no-VRP" benchmark. In the free-omega model (A4f), VRP correction is distributed across two channels—*θ_1 *scaling and* $\mathbb{E}[g_t]$ *level—so* θ_1 *alone no longer identifies the full VRP discount; see the empirical discussion below.*

Empirically, the constrained model yields θ_1-ratio $= 0.78$, implying a 21.9% discount—directly measuring the average VRP fraction. In the free-omega model, VRP correction splits between two channels: θ_1-ratio $= 1.96$ (overshoot) and $\mathbb{E}[g_t] = 0.48$ (level correction), with effective ratio $= 0.94$. This demonstrates that VRP correction is a real economic phenomenon captured by the multiplicative structure.

Remark 1 (g Tracks VRP Dynamics)**.** *The dynamic factor* g_t *reflects time-varying departures of realized variance from the VRP-corrected implied level. The GARCH autoregressive structure smooths the noisy daily ratio* r_t^2/τ_t*, extracting its persistent VRP component. Unlike Propositions 1 and 2, this characterization is empirical rather than derived analytically; it summarizes the decomposition behavior observed across the full sample.*

By construction, g_t from the model recursion is approximately orthogonal to VRP ($\rho \approx 0.06$), because τ_t already absorbs the VRP signal. The high correlation reported in Section 5.2 ($\rho \approx 0.80$) uses the g-proxy method of comparing model variance to raw implied variance without VRP correction, where the GARCH dynamics amplify the systematic VRP component relative to the raw ratio ($\rho = 0.15$).

These results distinguish source decomposition from mere relabeling: (i) θ_1 provides parametric identification of VRP magnitude (in the constrained model) or distributes

VRP correction across two channels (in A4f); (ii) the multiplicative structure provides structural scale separation—τ_t captures exogenous option-market fear while g_t captures endogenous GARCH dynamics, with this decomposition interpretable whether $\mathbb{E}[g_t] = 1$ (constrained) or $\mathbb{E}[g_t] = 0.48$ (A4f); and (iii) the forecasting gain (DM $t = 4.03$ vs. GJR) demonstrates that the decomposition captures information not available to standard models.

8 Conclusion

This paper provides a systematic evaluation of multiplicative volatility models that incorporate VIX as an external scale factor. Our main findings are as follows.

First, A4f and a majority—but not all—of the VIX-augmented multiplicative models significantly outperform plain GJR-GARCH under the $|t| > 3$ screen (A4f: DM $t = 4.03$). The Model Confidence Set eliminates only GJR at $\alpha = 0.25$, a set-level result consistent with useful VIX information but not a pairwise win for every specification. Among VIX-augmented models, A4f ($\tau_t = \theta_0 + \theta_1 \mathrm{VIX}^2_{t-1}$, free ω_g) achieves the lowest QLIKE, and significantly outperforms longer-lag MIDAS specifications (B2: $|t| = 3.04$; B3: $|t| = 3.73$), but is statistically indistinguishable from the shortest-lag MIDAS (B1, $K = 22$). The practical implication is that longer-lag GARCH-MIDAS specifications ($K \geq 65$) are significantly dominated by A4f ($|t| > 3.0$), while the shortest-lag MIDAS (B1, $K = 22$) achieves statistically indistinguishable performance (DM $t = -0.90$)—parsimony rather than statistical dominance is the practical argument for the daily specification.

Second, the VIX-squared functional form for τ_t dominates alternatives due to dimensional consistency: VIX^2 has variance units matching σ_t^2. The free-omega treatment accommodates the systematic wedge between implied and realized variance (the VRP), and the contemporaneous normalization ($u_{t-1} = r_{t-1}/\sqrt{\tau_t}$) outperforms the lagged alternative.

Third, the g_t component provides a structured decomposition of volatility into exogenous (option-market fear) and endogenous (GARCH dynamics) sources, with the latter tracking VRP dynamics at $\rho \approx 0.80$. However, g_t has no exploitable out-of-sample predictive power for future VRP, making it a contemporaneous decomposition tool rather than a forecasting signal.

Fourth, A4f achieves nominal Harvey significance for five of seven tested markets. Four use US VIX—SPY, QQQ, EURO STOXX 50, and FEZ—while the significant GLD specification uses its asset-specific GVZ index. EEM and 0050.TW fail to reach significance with US VIX, so the evidence supports cross-market usefulness only when the implied-volatility input is relevant to the asset.

Fifth, residual diagnostics reveal the mechanism behind A4f's improved VaR calibration: the VIX component absorbs extreme volatility episodes that GJR-GARCH must accommodate solely through fat-tailed innovations. A4f's standardized residuals exhibit 60% lower excess kurtosis (1.24 vs. 3.07) and a shift in optimal Student-t degrees of freedom from 5.3 to 8.0, resulting in a 76% reduction in the Jarque-Bera statistic. This "tail normalization" effect directly translates to VaR improvement: A4f-t achieves a scorecard of 3/4, substantially outperforming GJR-GARCH (1/4).

Our findings have implications for both researchers and practitioners. For researchers, the results suggest that model complexity should be matched to data frequency: component models with mixed-frequency aggregation are valuable when the external information is genuinely lower-frequency (e.g., quarterly GDP), but a simple daily specification is preferable when the variable updates daily. For practitioners, the A4f-t model offers a parsimonious, interpretable, and well-performing alternative to both standard GARCH and complex GARCH-MIDAS specifications for equity volatility forecasting and risk management.

Several limitations should be noted. First, our QLIKE evaluation uses r_t^2 as the volatility proxy. The proxy-robustness result of Patton (2011) concerns expected loss when the proxy satisfies the required conditional-unbiasedness assumptions; it does not guarantee that arbitrary proxy noise preserves a realized finite-sample ranking. Both the *magnitudes* of QLIKE differences and DM statistics, and potentially their finite-sample ordering, may differ when 5-minute realized variance is used as the proxy. Since VIX and intraday realized variance have a non-trivial relationship, the relative advantage of VIX-based models could change with proxy choice; we leave this robustness check for future work with high-frequency data. Second, our cross-asset validation covers six additional assets (seven including primary SPY); a broader study spanning international equity indices, fixed income, and commodity markets would strengthen the generalization claims. The VaR backtesting reveals persistent difficulty at the 2.5% level (DQ $p = 0.006$ for A4f-t), suggesting that violation clustering remains incompletely addressed. Finally, the free-omega A4f model adds one parameter relative to the constrained A4; while this improves fit for SPY and QQQ, the cross-asset comparison shows no significant A4f vs. A4 difference ($t < 1.7$ for all non-SPY assets), suggesting that the free omega is primarily useful for assets with strong VIX linkage.

Future research directions include incorporating realized variance measures from high-frequency data as the target variable, extending the multiplicative framework to multivariate settings for portfolio risk management (preliminary DCC-A4f results suggest significant gains over DCC-GJR for SPY/GLD portfolios), multi-horizon VaR calibration at $h = 5$ and 10 day horizons (where initial tests suggest $\sqrt{h}$ scaling provides adequate approximation to proper h-step GARCH formulas), and investigating whether the source decomposition can be exploited in longer-horizon VRP prediction with monthly or quarterly aggregation.

A Specification Genealogy and Multiple-Testing Disclosure

To address potential concerns about data mining and post-hoc specification selection, Table 14 documents the ex-ante design rationale for all 17 specifications in the horse race. All specifications were defined prior to examining any OOS QLIKE rankings; the selection exercise is purely an ex-post ranking of pre-registered designs.

Table 14: Specification genealogy: ex-ante design rationale for all 17 models

Spec	Category	Ex-ante design rationale
A1	GARCH-X	Inherited from an earlier pilot specification; included *only* to document the cost of estimation/forecasting inconsistency (τ_t in estimation, τ_{t-1} in OOS)
A2	GARCH-X	Log-exponential: convention from the GARCH-MIDAS literature (Engle et al., 2013)
A3	GARCH-X	Log-exponential with lagged denominator (τ_{t-1}): conservative timing variant
A4	GARCH-X	VIX-squared: dimensional consistency (VIX^2 has variance units matching σ_t^2)
A5	GARCH-X	VIX-level $\exp(\theta_1\,\mathrm{VIX})$: alternative scaling, included for completeness
A2f	GARCH-X	Log-exponential with free ω_g: accommodates the VRP systematic wedge
A4f	GARCH-X	VIX-squared with free ω_g: combines A4 dimensional consistency with VRP accommodation
A3f	GARCH-X	Log-exponential, lagged denominator, free ω_g: conservative timing + VRP flexibility
A2n	GARCH-X	Log-exponential with sample-mean normalization: alternative $\mathbb{E}[g_t] = 1$ enforcement method
A4n	GARCH-X	VIX-squared with sample-mean normalization: dimensional consistency + alternative normalization
B1	MIDAS-RW	Rolling MIDAS, $K = 22$ daily lags (≈ 1 month): shortest conventional lag (Engle et al., 2013)
B2	MIDAS-RW	Rolling MIDAS, $K = 65$ daily lags ($\approx$1 quarter): medium lag
B3	MIDAS-RW	Rolling MIDAS, $K = 125$ daily lags (≈ 6

References

Acerbi, C. and Szekely, B. (2019). Backtesting expected shortfall: Accounting for tail risk. *Management Science*, 65(12):5542–5567. https://doi.org/10.1287/mnsc.2018.3159

Bekaert, G. and Hoerova, M. (2014). The VIX, the variance premium and stock market volatility. *Journal of Econometrics*, 183(2):181–192.

Bollerslev, T. (1986). Generalized autoregressive conditional heteroskedasticity. *Journal of Econometrics*, 31(3):307–327. https://doi.org/10.1016/0304-4076(86)90063-1

Diebold, F.X. and Mariano, R.S. (2002). Comparing predictive accuracy. *Journal of Business & Economic Statistics*, 20(1):134–144.

Engle, R.F. (1982). Autoregressive conditional heteroscedasticity with estimates of the variance of United Kingdom inflation. *Econometrica*, 50(4):987–1007. https://doi.org/10.2307/1912773

Bollerslev, T., Tauchen, G., and Zhou, H. (2009). Expected stock returns and variance risk premia. *Review of Financial Studies*, 22(11):4463–4492.

Campbell, J.Y. and Thompson, S.B. (2008). Predicting excess stock returns out of sample: Can anything beat the historical average? *Review of Financial Studies*, 21(4):1509–1531.

Carr, P. and Wu, L. (2009). Variance risk premiums. *Review of Financial Studies*, 22(3):1311–1341.

Christensen, B.J. and Prabhala, N.R. (1998). The relation between implied and realized volatility. *Journal of Financial Economics*, 50(2):125–150.

Christoffersen, P.F. (1998). Evaluating interval forecasts. *International Economic Review*, 39(4):841–862.

Conrad, C. and Kleen, O. (2020). Two are better than one: Volatility forecasting using multiplicative component GARCH-MIDAS models. *Journal of Applied Econometrics*, 35(1):19–45. https://doi.org/10.1002/jae.2742

Conrad, C. and Loch, K. (2015). Anticipating long-term stock market volatility. *Journal of Applied Econometrics*, 30(7):1090–1114. https://doi.org/10.1002/jae.2404

Corsi, F. (2009). A simple approximate long-memory model of realized volatility. *Journal of Financial Econometrics*, 7(2):174–196. https://doi.org/10.1093/jjfinec/nbp001

Francq, C. and Thieu, L.Q. (2019). QML inference for volatility models with covariates. *Econometric Theory*, 35(1):37–72. https://doi.org/10.1017/S0266466617000512

Giacomini, R. and White, H. (2006). Tests of conditional predictive ability. *Econometrica*, 74(6):1545–1578.

Engle, R.F. and Manganelli, S. (2004). CAViaR: Conditional autoregressive value at risk by regression quantiles. *Journal of Business & Economic Statistics*, 22(4):367–381.

Engle, R.F. and Rangel, J.G. (2008). The spline-GARCH model for low-frequency volatility and its global macroeconomic causes. *Review of Financial Studies*, 21(3):1187–1222.

Engle, R.F., Ghysels, E., and Sohn, B. (2013). Stock market volatility and macroeconomic fundamentals. *Review of Economics and Statistics*, 95(3):776–797.

Glosten, L.R., Jagannathan, R., and Runkle, D.E. (1993). On the relation between the expected value and the volatility of the nominal excess return on stocks. *Journal of Finance*, 48(5):1779–1801. https://doi.org/10.1111/j.1540-6261.1993.tb05128.x

Han, H. and Kristensen, D. (2014). Asymptotic theory for the QMLE in GARCH-X models with stationary and nonstationary covariates. *Journal of Business & Economic Statistics*, 32(3):416–429. https://doi.org/10.1080/07350015.2014.897954

Hansen, P.R., Lunde, A., and Nason, J.M. (2011). The model confidence set. *Econometrica*, 79(2):453–497.

Harvey, C.R., Liu, Y., and Zhu, H. (2016). ...and the cross-section of expected returns. *Review of Financial Studies*, 29(1):5–68.

Newey, W.K. and West, K.D. (1987). A simple, positive semi-definite, heteroskedasticity and autocorrelation consistent covariance matrix. *Econometrica*, 55(3):703–708.

Jiang, G.J. and Tian, Y.S. (2005). The model-free implied volatility and its information content. *Review of Financial Studies*, 18(4):1305–1342.

Kupiec, P.H. (1995). Techniques for verifying the accuracy of risk measurement models. *Journal of Derivatives*, 3(2):73–84. https://doi.org/10.3905/jod.1995.407942

Patton, A.J. (2011). Volatility forecast comparison using imperfect volatility proxies. *Journal of Econometrics*, 160(1):246–256.

Wang, F. and Ghysels, E. (2015). Econometric analysis of volatility component models. *Econometric Theory*, 31(2):362–393.

White, H. (2000). A reality check for data snooping. *Econometrica*, 68(5):1097–1126. https://doi.org/10.1111/1468-0262.00152

Lai, Y.-H. (2026). Volatility targeting as drawdown insurance: Evidence from Taiwan. Working paper, Da-Yeh University.